Intro: Building Healthcare AI Assistant

This Project provides hands-on experience in building an end-to-end healthcare AI assistant using Microsoft Foundry. The solution is designed to support hospital patient-services teams by answering routine questions, retrieving appointment information, processing discharge documents, and generating safe, grounded responses. It combines generative AI, RAG, function calling, document understanding, multi-agent orchestration, and MCP integration within a single practical workflow.

👉 Want the full picture first? Check out the presentation for a visual walkthrough of the project.

What We're Building

In this lab guide, you'll build a comprehensive, production-minded healthcare AI assistant that demonstrates the full power of Microsoft Foundry. This isn't just a simple chatbot, it's a multi-agent system that can look up patients and appointments, ground its answers in real policy documents, read scanned medical paperwork, and reach outside knowledge safely, all while following hard-coded safety rules that never bend.

Business Scenario

You work for Northfield General Hospital, a fictional 300-bed regional hospital whose patient-services desk fields hundreds of calls a day.

Patients and families contact the desk for various reasons:

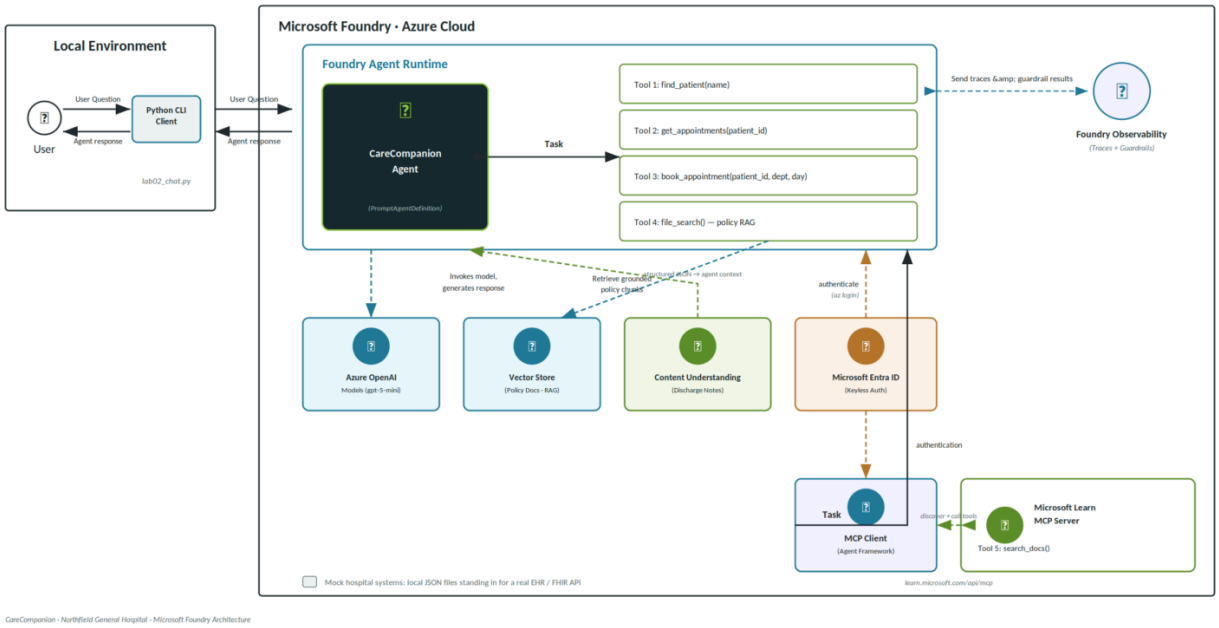
Logistics Questions: Visiting hours, appointment lookups, billing, directions, and parking

Discharge Confusion: Patients who were just discharged and can't remember what their paperwork said

Currently, staff spend a significant amount of time answering the same routine questions, leaving less time for the complicated cases that need real judgment. The hospital needs an AI solution that handles the routine work reliably, with hard rules against giving medical advice and for escalating anything that sounds like an emergency — while human staff stay in the loop for anything consequential.

Architecture Overview

CareCompanion represents a realistic hospital-facing application that showcases every major Foundry capability, a Foundry Agent Service agent, function tools over mock hospital data, managed RAG, Content Understanding, multi-agent orchestration, and MCP, inside one integrated architecture, all running from a single Foundry project.



Guide Structure

This guide follows an incremental enhancement approach, where each of the seven labs builds on the one before it starting from a first model call and ending with a monitored, multi-agent system ready for cleanup.

Total Guide Time: ~5–6 hours

By completing this guide, you will master building and operating a healthcare-ready AI agent using Microsoft Foundry to handle real hospital logistics, grounded knowledge, document understanding, and safe multi-agent orchestration.